Kanak Raj

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Education

- **Birla Institute of Technology Mesra** — Ranchi, India
  Integrated Master's of Science - Mathematics & Computing; GPA: 8.84 — 2019 - 2024

Publications

- Test-Time Scaling in the Wild: Why Exploitation, Not Exploration, Is the Bottleneck (joint first author) **at NeurIPS 2026.**
- Transformers Remember First, Forget Last: Dual-Process Interference in LLMs, arXiv 2025; extended version **under review at AAAI 2027.**
- K-PERM: Personalized Response Generation Using Dynamic Knowledge Retrieval and Persona-Adaptive Queries **at AAAI Spring Symposium 2024.**
- Multimodal Ingredient Substitution Knowledge Graph (MISKG) **Dataset Release 2024.**
- Cook-Gen: Robust Generative Modeling of Cooking Actions from Recipes **at IEEE SMC 2023.**
- Spatial Field Fusion Network (SFFNet) for Panoramic Dental X-ray Segmentation **at IEEE APSCON 2023.**

Ongoing Projects

- **Metacognitive Confidence in LLMs** — Using probing, activation steering, and patching, found sycophancy is encoded intrinsically at the same position that carries the model's confidence signal, prior to verbalization.

Experience

- **NielsenIQ** — Bangalore, India
  Research Scientist — Jul 2026 - Present
  - Agent trajectory optimization and setting up RL environments for SLM fine-tuning on custom tasks

- **Thomson Reuters Lab** — Bangalore, India
  Applied Research Scientist — Aug 2024 - Jun 2026
  - **LLM Reasoning & Reward Modeling:**
    - Fine-tuned Qwen3.5-9B via Dr. GRPO for multi-turn legal citation retrieval (1,139 Swiss law queries): designed 3-tool agentic environment (BM25 search, article verification, submit) through 29-experiment ablation — 7-tool design collapsed to F1 0.066 vs. F1 0.244 with 3 tools
    - Designed multi-stage reward: bootstrap stage rewards search behavior and citation format; optimize stage uses F1 with query quality, anti-hallucination, and over-prediction penalties
    - Built IRAC (Issue-Rule-Application-Conclusion) Knowledge Graph from Westlaw corpus for DPO fine-tuning preference data; improved LegalBench benchmark performance
  - **Test-Time Scaling:**
    - Implemented production module supporting Best-of-N, DVTS, Beam Search, and Particle Filtering steered by fine-tuned Bradley-Terry reward models (outcome and process)
    - Conducted the first systematic study of test-time scaling on non-verifiable tasks (NeurIPS 2026): compute-matched (inference-FLOPs) benchmarking of 5 TTS paradigms across 5 open-ended QA domains (legal, medical, finance, instruction-following, writing); ∼8,500 samples spanning deterministic to rubric-only verifiability
  - **Agentic Systems:**
    - Architected multi-agent system for automated legal document editing with error-resolving agents and multi-turn reasoning; achieved 98% accuracy, 65% reduction in manual revision time, sub-8s latency per section across hundreds of documents monthly

- **Microsoft Research** — Bangalore, India
  Research Intern (LLMs & AI4Code) — Jan 2024 - Jul 2024
  - Built PwR Studio: NL→DSL translation system enforcing symbolic compilation checks on LLM outputs; reduced hallucinations by 85%, valid code generation 65%→95%; featured in Satya Nadella's keynote, adopted by 15+ organizations
  - Architected Jugalbandi platform converting DSL into FSM-based chatbot applications; 80% reduction in development time; enabled government agencies and NGOs to deploy multilingual voice/text bots across WhatsApp, Telegram, and web; adopted by Bhashini for national chatbot infrastructure
  - Open-sourced: PwR-Studio, PwR-NL2DSL, Jugalbandi Studio Engine

- **AI Institute, University of South Carolina** — Remote
  Research Intern (Neuro-Symbolic AI & NLP) — Dec 2022 - Apr 2024
  - Master's thesis with Dr. Amit Sheth: built Multimodal Ingredient Substitution Knowledge Graph (MISKG) — 27K ingredients, 80K substitution pairs, sourced from FlavorDB, ConceptNet, and Recipe1M; largest ingredient substitution corpus; adopted at UC Irvine + Stanford Health Hackathon 2024
  - Formulated cross-modal recipe retrieval and cooking action recognition; 95% recall; led to Cook-Gen (SMC'23)